Prognostic Value of Neutrophil-to-Lymphocyte Ratio in Predicting the 6-Month Outcome of Patients with Traumatic Brain Injury: A Retrospective Study

Jian-Lan Zhao[1], Zhuo-Ying Du[1], Qiang Yuan[1], Jian Yu[1,2], Yi-Rui Sun[1,3], Xing Wu[1], Zhi-Qi Li[1], Xue-Hai Wu[1], Jin Hu[1]

BACKGROUND: **Peripheral white blood cells are regularly analyzed on admission for patients with traumatic brain injury (TBI). The prognostic value of the neutrophil-to-lymphocyte ratio (NLR) in predicting the 6-month outcome of patients with TBI is unclear.**

METHODS: **We designed a single-center retrospective cohort study. Patients admitted to Fudan University Huashan Hospital within 6 hours after TBI were identified between December 2004 and December 2017. The primary outcome was 6-month Glasgow Outcome Scale score. Independent predictors of 6-month outcome were assessed using uni- and multivariate analyses. Three models based on admission characteristics were built to evaluate the prognostic value of the NLR in predicting the outcome of patients with TBI. The discriminative ability of predictive models was evaluated by the area under the curve (AUC).**

RESULTS: **A total of 1291 patients with TBI were included. Multivariate analysis showed age, Glasgow Coma Scale scores at admission, subdural hematoma, intraparenchymal hemorrhage, traumatic subarachnoid hemorrhage, NLR (P < 0.001), and coagulopathy (P = 0.028) were independent predictors of 6-month outcome. The model combining the NLR and standard variables (AUC = 0.936; 95% confidence interval [CI], 0.923–0.949) was more favorable in predicting 6-month outcome of patients with TBI than the model without the NLR (AUC = 0.901; 95% CI, 0.883–0.919) and the model based only on the NLR (AUC = 0.827; 95% CI, 0.802–0.852).**

CONCLUSIONS: **NLR is an independent prognostic factor of predicting 6-month outcome of patients with TBI. A high NLR in patients with TBI is associated with poor outcome. The prognostic value of the NLR in predicting 6-month outcome of patients with TBI is favorable.**

INTRODUCTION

Traumatic brain injury (TBI) remains the leading cause of death and disability in young and adult patients.[1] The inflammatory response is one of the secondary injuries occurring soon after brain trauma. The mechanism of inflammatory injury is still not fully understood.[2] Neutrophils and astrocytes, microglia, and monocytes take part in the acute cellular reactions partly activated by purinergic signaling after TBI.[3] Studies have shown that neutrophils are recruited to the site of injury within 1 hour of focal TBI[4,5] and are capable of releasing inflammatory mediators, which induce neuronal death and secondary brain damage.[6-8]

A series of factors on patient admission have been shown to be prognostic of the outcome of patients with TBI, including international normalized ratio, activated partial thromboplastin time, platelet count, age, absence of pupillary reactivity, Glasgow Coma

Key words
- 6-Month outcome
- Neutrophil-to-lymphocyte ratio
- Prediction model
- Traumatic brain injury

Abbreviations and Acronyms
AUC: Area under the curve
CI: Confidence interval
CT: Computed tomography
GCS: Glasgow Coma Scale
GOS: Glasgow Outcome Scale
ICH: Intracerebral hemorrhage
IPH: Intraparenchymal hemorrhage
NLR: Neutrophil-to-lymphocyte ratio
OR: Odds ratio
SDH: Subdural hematoma
TBI: Traumatic brain injury
tSAH: Traumatic subarachnoid hemorrhage
WBC: White blood cell

From the [1]Department of Neurosurgery, Huashan Hospital, Fudan University, Shanghai; [2]Department of Neurosurgery, Shigatse People's Hospital, Shigatse, Tibet Autonomous Region; and [3]Department of Neurosurgery, Minhang Branch, Zhongshan Hospital, Fudan University, Shanghai, P.R. China

To whom correspondence should be addressed: Jin Hu, M.D., Ph.D. [E-mail: 286644104@qq.com]

Jian-Lan Zhao, Zhuo-Ying Du, and Qiang Yuan contributed equally to this work.

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Scale (GCS) score, and so forth.[9,10] Different types of predictive models have been built based on these prognostic factors; however, their prognostic values are still limited. Therefore, a model with higher prognostic value to predict the outcome of patients with TBI is required.

It has been reported that higher neutrophils and lower lymphocytes at admission are independently associated with poor outcome of patients with intracerebral hemorrhage (ICH), and the neutrophil-to-lymphocyte ratio (NLR) represents a readily available prognostic predictor.[11-13] However, the prognostic value of the NLR alone and in combination with other prognostic factors for the outcome of patients with TBI is unclear.

The aim of our study is to evaluate the prognostic value of the NLR and to test its predictive power in prediction models for the 6-month outcome of patients with TBI.

METHODS AND PATIENTS

Patient Population

A retrospective, observational study was performed. Patients with TBI who were admitted to the Department of the Neurosurgery Neurotrauma Center at Fudan University Huashan Hospital between December 2004 and December 2017 were included. This study was approved by the Ethical Review Boards of Fudan University Huashan Hospital. Informed consent was obtained from all individual participants. If a patient could not sign informed consent by himself/herself, such as patients with a Glasgow Outcome Scale (GOS) score of 1, informed consent was signed by his/her statutory guardian. This article is reported following the STrengthening the Reporting of OBservational studies in Epidemiology reporting guidelines.

Inclusion criteria are as follows. First, participants had to be computed tomography (CT) scan—confirmed patients with TBI. CT signs of TBI included epidural hematoma, subdural hematoma (SDH), intraparenchymal hemorrhage (IPH), brain contusion, and subarachnoid hemorrhage. Second, patients had to be ≥14 years of age. Finally, participants had to be admitted within 6 hours after injury. Patients with TBI with traumatic injury to a body region other than the brain with an Abbreviated Injury Severity score ≥3 and those with penetrating brain injury were excluded from the study. All patients were evaluated and treated by full-time neurosurgeons with specific training in critical care following guidelines for the management of severe traumatic brain injury.[14,15]

Demographic Data and White Blood Cell Counts

Baseline characteristics, including age, sex, mechanism of injury, pupillary reaction to light, GCS score at admission, and type of injury, were recorded for all patients. Injury types were assessed based on initial CT scan on admission. The 6-month outcome of patients with TBI was assessed using the GOS through outpatient interviews or over the telephone. GOS score of 1–3 was considered a poor outcome, and GOS score of 4 or 5 was considered a good outcome. Coagulopathy was defined as regular coagulation test results meeting one or more of the following criteria: platelet count $<100 \times 10^9$/L, international normalized ratio >1.25, prothrombin time >14 seconds, and activated partial thromboplastin time >36 seconds. Peripheral blood analysis was performed for all patients within 6 hours of injury at the Central Clinical Laboratory in Huashan Hospital. White blood cell (WBC) count, neutrophil ratio, lymphocyte ratio, and NLR were recorded.

Prediction Models

The predictors for 6-month outcome were first analyzed using the χ^2 test, Fisher exact test, t test, or 1-way analysis of variance. Then, a multiple logistic regression analysis was performed to further estimate the associations between predictors and outcome. Three models for 6-month outcome prediction were developed based on baseline characteristics: model A consisted only of standard predictors such as age, sex, mechanism of injury, pupillary reaction to light, GCS score at admission, and type of injury; model B further included the results of WBC count, neutrophil ratio, lymphocyte ratio, and NLR in addition to the predictors from model A; and model C contained only results of WBC count, neutrophil ratio, lymphocyte ratio, and NLR.

Statistical Analysis

Continuous variables were expressed as mean ± SD or median (interquartile range), and categorical variables were expressed as percentages. The univariate analyses of categorical data were performed using the χ^2 test. Equality of variance was assessed using the Levene test. Normally distributed variables were compared using Student t test or 1-way analysis of variance, whereas nonnormally distributed variables were compared using the Kruskal-Wallis or Mann-Whitney U test.

After the univariate analyses, a forward stepwise logistic regression analysis of the 6-month outcome was used to develop the prediction models and adjust for multiple predictors of 6-month outcome. By constructing the receiver operating characteristic curve and calculating the area under the curve (AUC), we evaluated the specificity and sensitivity of the 3 models, and their discriminative ability. All statistical tests were 2-tailed, and $P < 0.05$ was considered statistically significant. Statistical analysis was carried out using SPSS 23.0 (IBM, Armonk, New York, USA) and MedCalc statistical software (version 15.2.2 [MedCalc Software bvba, Ostend, Belgium]).

RESULTS

A total of 1291 patients with TBI were included in our study. Among these patients, 341 patients (26.42%) had poor outcome at 6 months after injury.

Univariate analysis revealed that age, pupillary reaction (both reacting), GCS score at admission, some types of injury (SDH, epidural hematoma, IPH, traumatic subarachnoid hemorrhage [tSAH], and diffusion axonal injury), and coagulopathy were significantly related to 6-month outcome ($P < 0.001$). Meanwhile, WBC count, neutrophil ratio, lymphocyte ratio, and NLR all were also closely correlated with the outcome of patients with TBI ($P < 0.001$). Patients with poor outcome had a significantly higher WBC count, neutrophil ratio, and NLR and a lower lymphocyte ratio than those with good outcome (Table 1).

Multivariate analysis was further performed using forward stepwise logistic regression to identify the prognostic factors of 6-month outcome. The result revealed that the NLR was an independent prognostic factor of 6-month outcome of patients with TBI after adjustment (odds ratio [OR], 0.91; 95% confidence interval [CI],

Table 1. Baseline Characteristics According to the 6-Month Outcome

Characteristic	Full Cohort (N = 1291)	Poor Outcome (GOS Score 1–3) (n = 341)	Good Outcome (GOS Score 4–5) (n = 950)	*P* Value
Age (years)	47.03 ± 16.88	50.98 ± 16.37	45.62 ± 16.85	<0.001
Male	982 (76.1)	257 (75.4)	725 (76.3)	0.725
Mechanism of injury				0.191
Motor vehicle accident	705 (54.6)	196 (57.5)	509 (53.6)	
Fall	154 (11.9)	44 (12.9)	110 (11.6)	
Stumble	205 (15.9)	40 (11.7)	165 (17.4)	
Blow to head	100 (7.7)	27 (7.9)	73 (7.7)	
Others	127 (9.8)	34 (10.0)	93 (9.8)	
Pupillary reactions				
Both reacting	1135 (87.9)	236 (69.2)	899 (94.6)	<0.001
One reacting	112 (8.7)	71 (20.8)	41 (4.3)	
None reacting	44 (3.4)	34 (10.0)	10 (1.1)	
GCS score at admission	11.21 ± 3.70	7.37 ± 2.96	12.59 ± 2.88	<0.001
Type of injury				
SDH	378 (29.3)	160 (46.9)	218 (22.9)	<0.001
EDH	368 (28.5)	76 (22.3)	292 (30.7)	0.003
IPH	860 (66.6)	286 (83.9)	574 (60.4)	<0.001
tSAH	649 (50.3)	222 (65.1)	427 (44.9)	<0.001
DAI	46 (3.6)	32 (9.4)	14 (1.5)	<0.001
Skull fracture	299 (23.2)	67 (19.6)	232 (24.4)	0.073
Coagulopathy	317 (24.6)	101 (29.6)	216 (22.7)	0.013
WBC count ($\times 10^9$/L)	13.68 ± 4.89	17.00 ± 4.97	12.48 ± 4.27	<0.001
Neutrophil ratio (%)	0.83 ± 0.08	0.88 ± 0.05	0.80 ± 0.07	<0.001
Lymphocyte ratio (%)	0.12 ± 0.06	0.07 ± 0.05	0.14 ± 0.06	<0.001
NLR (%)	12.18 ± 10.89	24.71 ± 12.52	7.68 ± 6.54	<0.001

Data are given as mean ± SD, number of patients (%), or as otherwise noted.

GOS, Glasgow Outcome Scale; GCS, Glasgow Coma Scale; SDH, subdural hematoma; EDH, epidural hematoma; IPH, intraparenchymal hemorrhage; tSAH, traumatic subarachnoid hemorrhage; DAI, diffusion axonal injury; WBC, white blood cell; NLR, neutrophil-to-lymphocyte ratio.

0.89–0.93; $P < 0.001$), whereas WBC count and neutrophil or lymphocyte ratio all showed no close relationship with patients' 6-month outcome (Table 2). Other independent prognostic factors included age, GCS score at admission, SDH, IPH, tSAH ($P < 0.001$), and coagulopathy ($P = 0.028$) (Table 3).

Three predictive models for the 6-month outcome were built. Their receiver operating characteristic curve and AUC were calculated to assess their discriminative ability (Figure 1). The model combining the NLR and other prognostic factors (AUC = 0.936; 95% CI, 0.923–0.949) showed more favorable discriminative ability than the model without the NLR (AUC = 0.901; 95% CI, 0.883–0.919) and the model using the NLR only (AUC = 0.827; 95% CI, 0.802–0.852). Therefore, using the NLR in addition to the standard prognostic factors can further improve the predictive ability of the prediction model for 6-month outcome of patients with TBI.

DISCUSSION

In this study, we investigated the prognostic value of the NLR on admission and developed prognostic models to predict 6-month outcome of patients with TBI. The result showed that a higher NLR was associated with a lower GOS score in patients with TBI at 6 months after injury. NLR was an independent prognostic factor of 6-month outcome of patients with TBI in both univariate and multivariate analysis. Combining the NLR with other standard prognostic factors can further improve the predictive power of the standard model for 6-month outcome after TBI.

To our knowledge, the prognostic value of the NLR has not been well studied. This study confirmed it an independent prognostic factor and its effect in improving predictive power when combined with the standard predictive model. Multivariate

Table 2. Associations of Leukocyte Count and Neutrophil-to-Lymphocyte Ratio with the 6- Month Outcome of Patients with Traumatic Brain Injury

	Unadjusted		Adjusted	
Independent Variable	**OR (95% CI)**	***P* Value**	**OR (95% CI)**	***P* Value**
WBC count ($\times 10^9$/L)	0.81 (0.79—0.84)	<0.001	1.04 (0.99—1.09)	0.156
Neutrophil ratio	0.80 (0.78—0.82)	<0.001	0.94 (0.89—1.04)	0.168
Lymphocyte ratio	1.28 (1.24—1.32)	<0.001	0.92 (0.84—1.02)	0.103
NLR	0.89 (0.87—0.91)	<0.001	0.91 (0.89—0.93)	<0.001

OR, odds ratio; CI, confidence interval; WBC, white blood cell; NLR, neutrophil-to-lymphocyte ratio.

analysis showed that age, GCS score at admission, SDH, IPH, tSAH, coagulopathy, and NLR were independently associated with 6-month outcome in patients with TBI. These predictive variables can be readily obtained on admission to a neurosurgical unit. Predictive models based on these prognostic predictors showed good discrimination compared with models based only on standard variables or the NLR as assessed by the AUC. Therefore, the combination of standard predictors and the NLR had better predictive power than standard predictors or the NLR alone.

Outcome prediction of patients with TBI on admission is crucial for both promotion of effective patient care and quality control within and across institutions.[16] The prognostic value of predictors is determined by their reliability on assessment, the prevalence of abnormalities, and the strength of the prognostic effect. In our research, the prognostic value of factors such as age, GCS score, and coagulopathy had been confirmed and widely accepted as a standard outcome predictor of patients with TBI.[10,17]

Peripheral blood analysis including WBC count, neutrophil ratio, lymphocyte ratio, and NLR, is a readily available, reliable, and standardized routine laboratory test across most medical centers. In neurosurgical patients, the importance of WBC count and its components in predicting outcome is increasingly recognized. NLR, which conveys crucial information about the complex inflammatory activity in the vascular bed,[18] is an established marker of systemic inflammation and is easily calculated. In ischemic diseases, the NLR is considered a predictor of poor outcome. However, the prognostic value of the NLR in predicting hemorrhagic diseases and TBI is not so clear. Favorable prognostic values of the neutrophil ratio, lymphocyte ratio, and NLR have been reported in patients with ICH in several previous studies.[11-13] Chen et al.[19] found that the neutrophil count was associated with early neurologic deterioration. Lymphocytes were reported to potentiate cerebral inflammation and brain injury.[20] A high NLR at admission was also reported to be an indicator of poor outcome in patients with ICH[19]; however, the specific underlying mechanism is still unclear. In our study, patients with TBI with poor outcome showed a significantly higher NLR than those with good outcome, which is consistent with previous studies. Moreover, the NLR of patients with TBI is consistently higher than that of patients with ICH. The possible reasons might be that ICH and TBI are different types of injuries by nature. Most ICHs are largely focal insult to the brain, whereas injuries from TBI are commonly more severe and diffuse.[21] As a result, more severe acute inflammatory responses may be induced in patients with TBI. As an indicator of the severity of acute inflammation, the level of the neutrophil ratio will increase greatly compared with other types of leukocytes after TBI. This explains the early increase of the NLR after injury and the much higher level of the NLR in patients with TBI compared with ICH.

Inflammation response is significantly associated with poor outcome of patients with TBI.[22] Such inflammatory response after TBI may initially be triggered by damage-associated molecular patterns released after brain tissue injury.[23,24] Because of early blood—brain barrier disruption, a more significant leukocyte infiltration can be observed compared with that in ICH,[25] which will further aggravate focal inflammatory response and lead to the worsening of secondary brain injury.[26] As reported by Kolaczkowska and Kubes,[27] the count of neutrophils increased, but the count of lymphocytes showed no significant changes in the acute phase of TBI. Therefore, the NLR increased greatly in patients with TBI at admission. Within the first hour, neutrophils are actively recruited around injured brain parenchyma and

Table 3. Multivariate Logistic Regression Analysis Predicting the 6-Month Outcome

Independent Variable	Adjusted OR (95% CI)	*P* Value
Age	0.98 (0.96—0.99)	<0.001
GCS score	1.60 (1.50—1.70)	<0.001
SDH	0.57 (0.39—0.84)	<0.001
IPH	0.45 (0.28—0.72)	<0.001
tSAH	0.66 (0.45—0.96)	<0.001
Coagulopathy	1.18 (1.01—1.39)	0.028
NLR	0.91 (0.89—0.93)	<0.001

OR, odds ratio; CI, confidence interval; GCS, Glasgow Coma Scale; SDH, subdural hematoma; IPH, intraparenchymal hemorrhage; tSAH, traumatic subarachnoid hemorrhage; NLR, neutrophil-to-lymphocyte ratio.

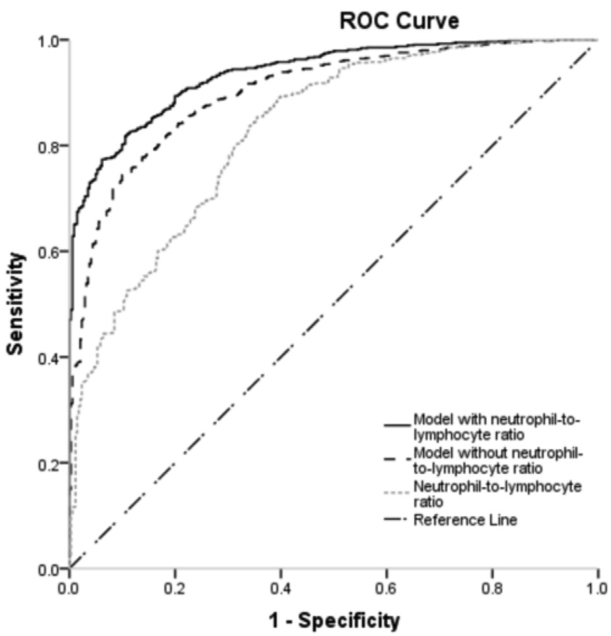

Figure 1. Receiver operating characteristic curve of the 3 predictive models. The model with the neutrophil-to-lymphocyte ratio (NLR) had a larger area under the curve than the other 2 models. It is indicated that the discrimination of the model with the NLR is more favorable than the other 2 models. ROC, receiver operating characteristic curve.

contribute to cellular injury and disruption of viable brain tissue in patients with TBI, which contributes as the secondary brain injury and leads to poor patient outcome.[7,28]

It is widely accepted that acute inflammatory responses are the main mechanism leading to secondary brain injury both in patients with ICH and TBI, and that infiltration of neutrophils into the site of hemorrhage contributes to focal inflammation and brain injury. Therefore, patients with TBI with a higher NLR are more likely to suffer from more severe secondary brain injury and have a poor 6-month outcome. In our study, the results of uni- and multivariate analysis showed that only the NLR was significantly associated with the outcome. It is revealed that WBC count, neutrophil ratio, and lymphocyte ratio cannot independently predict the 6-month outcome in patients with TBI with favorable prognostic value.

There several drawbacks in this study. First, the retrospective nature makes it vulnerable to biases. Second, being the largest neurotrauma center in the region, we tend to admit patients with more severe injuries, which can cause admission bias. Third, we know that inflammation and secondary brain injury are highly time sensitive. As a major referral center, we admit patients who are transported from as far as 6 hours' drive away after injury. Therefore, high heterogeneity may well exist among the NLR results taken from different time points after injury. Also, in further studies, we will include patients with isolated head injury to get a more reliable result. Therefore, caution should be exercised in interpreting our conclusions, and a prospective multicenter study is justified to further elucidate the potential mechanism of this phenotype.

CONCLUSIONS

NLR is an independent predictor of 6-month outcome in patients with TBI. The prognostic value of the NLR to 6-month outcome of patients with TBI is favorable. A high NLR in patients with TBI is associated with poor outcome.

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